

Manifesto for open agricultural robotics — Sustainable Robotics Base for Crops

Agricultural robotics needs a shared base that is open, lean, safe, and documented — one that can respect the diversity of farming practices. That is the purpose of Sustainable Robotics Base for Crops (SRBC): to build and share a production software foundation for autonomous agricultural robots, so that the value created by technology also returns to those who make it live — farmers — and strengthens a fair, maintainable, and durable supply chain.

We affirm that open agricultural robotics can help accelerate the adoption of agroecological practices by fostering the diffusion of effective, appropriable, and safe tools. Agronomic, ecological, economic, and social performance must become clearly measurable — and must stop being set against each other.

We call on farmers, developers, manufacturers, workshops, integrators, researchers, cooperatives, and educational institutions to join this effort: publish an interface, stabilize a format, write documentation, run a trial, train a neighbour. Nothing prevents us from innovating quickly and well; everything invites us to do it together.

Technology should not be a hindrance. It should be a bridge toward the durability of agricultural systems.

Why open

When we seeded the SABI AGRI project in 2015, the goal was already to diffuse agricultural equipment in the service of a more sustainable and more sovereign agriculture. Our experience as a manufacturer of electric machines and agricultural robots has confronted us with a structural reality: the speed of digital cycles and the closed nature of many solutions create a disproportionate dependency for the farming world.

On the one hand, the obsolescence of hardware and software platforms is out of step with agriculture's long timescales. On the other, proprietary interfaces lock users into an unbalanced relationship with their suppliers — who are themselves dependent on the technologies of their own suppliers. A machine that can no longer be understood, maintained, or adapted eventually becomes a constraint, even when it was initially performant.

We believe every farm should be able to benefit from technological advances without being captive to closed formats. To open is to make repair, adaptation, and the transmission of knowledge possible at the pace of seasons and territories.

For whom, and in the service of what

We situate the SRBC robotics ecosystem in the service of small and medium-sized farms, especially when they are diversified and engaged in agroecological practices. They play an essential role in ecosystem balance, the diversity of production, the vitality of territories, and the preservation of contextual know-how. Yet these farms are also particularly exposed to the hardship of physical work, labour needs, and barriers to investment.

Agroecological practices often require more observation, more precision, and more care for soils and living systems. Robotics can help make these practices economically accessible by reducing hardship and making operations favourable to crops, soils, and ecosystems feasible. Farmers' decisions can then take on their full importance in ecosystem management.

Not every form of robotics serves this ambition. Depending on its architecture and business model, it can reinforce the concentration of means of production and technological dependency. The issue is therefore not to adopt robotics as a matter of principle, but to determine which agricultural models it is designed to serve, who can access it, who understands it, who sets its uses, and who captures the value it creates.

What SRBC is

Sustainable Robotics Base for Crops designates the public ecosystem of software building blocks developed, among other things, for the SRBC robot and, more broadly, for interoperable agricultural robotics.

This is not a demonstration detached from the field. A substantial part of this code is used in production: navigation, localization, geofencing, perception, simulation, mission format, and associated tools. The repositories are published under the GitHub organization [Sustainable-Robotics-Base-for-Crops](#).

The SRBC ecosystem is a set of versioned packages and tools — mainly under ROS 2 — designed to be read, audited, reused, or replaced independently. Open contracts (the JSON Agri mission format, communication interfaces, navigation messages) allow third-party tools, integrators, and researchers to produce or consume missions without depending on the secrets of a single manufacturer.

We assume a clear boundary: openness primarily covers generic foundations and interfaces; industrial integration and the machine's own safety chain remain the manufacturer's responsibility. Opening a commons must go hand in hand with a clear chain of responsibility, while making shareable the knowledge needed for interoperability, maintenance, and collective innovation.

Proportionate, modular, open, and responsible design

For robotics to serve agroecology, its design choices must be coherent.

We favour machines that are proportionate to their uses: capable enough to deliver a real service, light and sober enough to remain accessible, maintainable, and compatible with care for the soil.

We favour modularity: separating the carrier, implements, sensors, and software functions so that systems can be repaired, adapted, retrofitted, and shared without rebuilding everything at every evolution.

We favour openness: public interfaces, documentation, tests, versioning, and licences that allow study, use, modification, and redistribution. Open source is not an end in itself. It is an industrial method for pooling foundations and preserving sovereignty of use over time.

We finally favour responsible innovation: anticipating social and environmental effects, involving users, documenting limits of use, governing agricultural data carefully, and maintaining explicit industrial responsibility for every machine placed in service.

These principles are not a label. They are a requirement of coherence: a technology is useful only if it durably increases farmers' capacity to understand, decide, and act.

What openness brings

In a fragmented sector, too much energy is spent re-implementing the same foundations. Pooling these building blocks does not destroy industrial activity: it relocates it.

For farmers, the stake is sovereignty of use: diagnosing, repairing, evolving systems, choosing a service provider, and retaining access to data. For manufacturers, differentiation shifts toward integration, robustness, demonstrated safety, and service. For workshops and integrators, documented interfaces make local multi-brand competence and territorial capillarity possible. For research and education, open formats finally make it possible to compare, teach, and transmit.

A technological commons needs documentation, maintenance, and governance. Openness is sustainable only if economic models — machines, tools, services, training — allow companies to thrive without recreating captivity by other means.

Even with open software code, safety, conformity, and responsibility remain attached to every configuration placed in service. An autonomous function only makes sense within an explicit operating domain, with predictable behaviour outside that domain and evidence of safe stop. We reject the naïve opposition between openness and industrialization: to industrialize is also to make quality, maintenance, interoperability, and safety reproducible.

Call to action

The diffusion of agricultural robotics in the service of agroecology is built with and among users, together with manufacturers and technical communities.

Everyone can contribute according to their skills: publish an interface, document a procedure, propose a compatible tool, improve a building block, host an experiment, train a neighbour, share field feedback together with its limits.

Opening agricultural robotics means refusing to let it become an instrument of dependency. It means giving the farming world the means to take part in its design, to master its uses, and to transmit its knowledge.

Join the ecosystem: github.com/Sustainable-Robotics-Base-for-Crops

About the author

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Since 2015, Alexandre Prévault-Osmani has worked at the intersection of electrification, agricultural robotics, open source, and agroecology. He develops an industrial approach aimed at making agricultural technologies accessible, interoperable, maintainable, and appropriable by the farming world.

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